Artificial Intelligence Use and Potential within Project Management

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Abstract. This study explores the impact of Artificial Intelligence (AI) on project management in Norway using a mixed-methods approach. Initially, a questionnaire was answered by 84 respondents, followed by in-depth interviews with seven experienced project managers. The primary objective was to assess the extent of AI integration in project management practices and its perceived benefits. The findings reveal that a significant number of project managers in Norway have already incorporated AI tools into their workflows. These tools are primarily utilized for task automation, predictive analytics, and resource allocation. Project managers reported that AI significantly enhances efficiency by automating routine tasks, thereby allowing them to focus on more strategic activities.
Additionally, AI tools provide valuable insights through data analysis, enabling more informed decision-making and greater project flexibility. Despite the overall positive reception, some respondents highlighted challenges such as the steep learning curve associated with new AI technologies and concerns over data security. These challenges, however, are manageable and can be addressed with proper training and robust data protection measures. The consensus is clear- AI has the potential to transform project management, making it more dynamic and adaptable to changing project demands. This study underscores the growing importance of AI in the project management domain in Norway. It suggests that ongoing advancements in AI technology will likely lead to even greater integration and reliance on these tools in the future. Further research is recommended to explore long-term impacts and the development of best practices for AI adoption in project management.

Keywords: Project management, Artificial intelligence, Agile.

1 Introduction

Artificial Intelligence (AI) is revolutionizing various industries, and project management is no exception. By leveraging advanced algorithms and machine learning, AI is transforming how projects are planned, executed, and monitored [3][24]. Traditionally, project management relied heavily on human judgment and manual processes, which often led to inefficiencies and errors. AI, however, offers tools and solutions that enhance precision, efficiency, and adaptability.

AI can automate routine tasks, such as scheduling, tracking progress, and managing resources, allowing project managers to focus on strategic decision-making [24]. Predictive analytics, powered by AI, enables more accurate forecasting of project timelines, budgets, and potential risks [1][33]. This proactive approach helps in identifying issues before they become critical, thereby improving project outcomes [33].

Moreover, AI-driven data analysis provides deeper insights into project performance, facilitating more informed decision-making. For instance, AI can analyze historical project data to identify patterns and trends, offering recommendations for optimizing current and future projects [29]. This capability not only enhances efficiency but also promotes greater flexibility in managing dynamic project environments.

As AI technology continues to evolve, its integration into project management practices is expected to grow, offering even more sophisticated tools and techniques. Table 1 shows examples of AI technologies supporting project management tasks. Understanding and harnessing the power of AI is becoming essential for project managers aiming to stay competitive and deliver successful projects in an increasingly complex and fast-paced world.

Table 1. AI technologies supporting project management tasks.

AI Technology	**Project Management Task**	**- Advantages**
Predictive analytics	Error detection	- Earlier recognition of possible errors. - Minimizing the risk. - Increase the standard of the project outcome.
Machine learning	Outcome forecasting	- Provide predictions about the project outcome. - Help in risk assessment and contingency planning.
Agile project assistant	Agile methodologies	- Enhance the agility of the project implementation. - Support risk assessment. - Contingency planning
Speech recognition software	Communication management	- Streamlines communication within teams and with stakeholders. - Enhance clarity and record-keeping
Natural language processing (NLP)	Documentation and reporting	- Automates the creation and analysis of project documents and reports. - Saving time and improving accuracy
Decision support systems	Strategic decision making	- Helps project managers make educated decisions by offering data-driven insights.
Robotic Process Automation (RPA)	Routine task automation	- Automates repetitive and routine tasks, freeing up human resources for more complex project issues.

The four variables discussed in this research are AI integration, organizational learning culture [4], change management practices [27], and project management agility. The discussion part explains if there is an impact of AI integration, organizational learning culture, and change management practices on project management agility.

1.1 Problem statement

There is a constant demand for project managers to balance conflicting priorities and handle several tasks at once. Project managers are expected to be efficient, and the use of AI tools may improve the efficiency and quality of project management. Nieto-Rodriguez and Vargas [29] observed that most organizations and project managers are still using spreadsheets, slides, and other applications that have not evolved much over the past few decades. In their opinion, these traditional tools fall short in an environment where projects and initiatives are constantly adapting and continuously changing the business. Bhavsar, Shah, and Gopalan [6] see AI as a potential game changer for the software project management and development life cycle processes. They say AI can help project managers focus on establishing organizational goals through cost optimization and improving the quality of the product.

Previous research regarding the use of AI in project management has frequently focused on project management issues concerning specific applications of AI [21], but specifically not in Norway. A low number of studies were conducted in 2007, 2008, 2010, and 2012, and there was a significant increase in AI-related studies in 2019 and 2020 [12]. Findings by Taboada et al. [32] indicated a significant rise during the time between 2013 – 2023 of AI use in project management, and Bley et al. [7] found there is a positive relationship between organizational culture, AI capabilities, and organizational performance. Engel, Ebel, and Giffen [15] examine the relationship between attributes of AI, project management, and organizational change. AI application in project management was discussed by Davahli [14], who evaluated 652 articles from several databases and concluded that there are still a lot of unrealized prospects in the early phase of AI use in project management [14].

Applications of AI have been considered and deployed to help project management procedures in Norway. However, a vast area of knowledge still needs to be investigated on how to incorporate AI technologies into project management practices. Previous research could propose concepts regarding AI adoption. However, there needs to be more empirical research concentrating on the operationalization of AI tools and methods within the Norwegian project management context. There is a clear need for research work that exhibits how AI could optimize project management performance, as well as assessing factors unique to the Scandinavian environment that influence its adoption and implementation within Norwegian organizations.

1.2 Research Objectives

Primary Objective

To investigate the utilization of AI in project management inside organizations based in Norway.

Secondary Objectives

- **Objective 1**: To find out whether there is an impact of AI integration on project management agility in Norwegian companies.
- **Objective 2**: To find out whether there is an impact of organizational learning culture on project management agility in Norwegian companies.

- **Objective 3**: To find out whether there is an impact of change management practices on project management agility in Norwegian companies.

1.3 Research Questions

Primary Research Questions
What is the extent of AI utilization in project management within organizations based in Norway?

Secondary Research Questions
- **RQ01**: Does AI integration impact project management agility in Norwegian companies?
- **RQ02**: Does organizational learning culture impact project management agility in Norwegian companies?
- **RQ03**: Does change management practices impact project management agility in Norwegian companies

2 Literature Review

This section presents a literature review. After presenting some background on AI, the discussion moves to project management and, more specifically, agile project management and ends up with a discussion of AI in project management.

2.1 Artificial Intelligence

AI is capable of advancing the degree of thinking in computer systems [28] and focuses on programming computers to perform tasks that people are currently more adaptable to [5]. AI can be described as a technology that can complete tasks that humans are unable to complete without cognitive ability [28]. The use of AI increases the possibility of computers performing tasks that are currently handled by people [23] while making dramatic shifts in company governance, organizational culture, and efficiency [8].

Researchers have been able to come up with various types of uses, including the detection of faces, picture identification, and more [36]. AI provides opportunities for massive data analysis and for doing routine work by identifying problems that humans might miss and providing creative and new solutions for the issues [36]. AI can assist in recognizing errors and alerting users when intervention is required, as well as it is capable of planning and scheduling resources based on the goals of the project [25].

There are some challenges associated with the usage of AI as well. According to Wang [36], the more complicated a function that a company needs to use, the more amount of data will be required, and it may lead to privacy concerns. The availability of data resources might also be an issue [36]. Sometimes, AI systems could provide inaccurate information owing to insufficient system training [25].

The improvement of AI may lead to unemployment; it may cause a drop in the reputation of a company [36]. Even though the use of AI is a really cost-effective way, there are some potential dangers associated with it, such as it not providing accurate results due to the inadequate amount of data for analysis, there may be restricted access to needed data, or not having enough knowledge to work with AI can be some problems. Another challenge of AI is that it can highly affect the future job market, and researchers have already studied how it will be affected; Fridgeirsson et al. [17] mentioned in their article that Frey and Osborne were the first to publish their investigations, which determined that approximately 47% of US employees are threatened by AI job automation.

Several studies have investigated bias regarding AI-integrated decision-making. For this process, algorithms are trained using the previous data so they can reinforce pre-existing biases when trained on historically biased data, which may lead to discriminatory practices [2]. As an example, in the process of recruiting employees, AI-powered apparatuses might accidentally incline toward candidates from certain statistic-bunches, sustaining systemic disparities [2].

2.2 Project Management

Project management is the application of necessary knowledge and skills to achieve a common goal, and it has been using AI since 1987 [24]. Nagireddy [28] stated the four primary elements of project management as time, cost, scope, and quality. The productive implementation of a project is dependent upon the abilities of project managers, assistants, and directors who ensure follow-up and evaluation [18].

According to Collins [12], many projects fail despite the use of project management approaches, and such failures limit the potential of technological components. Most projects fail because of late deliveries, over budget, and lack of planning. Additionally, project managers need to get adequate support from the top management within most organizations [6].

2.3 Agile Project Management

The Agile Project Management Model is an iterative and adaptive method that highlights mobility, cooperation, and adaptability to shifting needs, which closely correlates with the dynamic nature of today's corporate settings [31]. The main objective is to deliver the project outcomes through several sprints. Then, the project team can adapt according to the feedback of the customer [11]. The productive implementation of a project is dependent upon the abilities of project managers, assistants, and directors who ensure follow-up and evaluation [16]. Frameworks like Scrum, Kanban, and Extreme Programming (XP) are integral components of the Agile Project Management Model.

2.4 AI in Project Management

Shortly, technological developments are going to have a significant influence on project management, and AI will undoubtedly change how project management tasks are performed and handled. When considering the previous investigations, project managers spend over fifty percent of their days engaged in bureaucratic tasks, including improvements and monitoring [33]. According to research done by Fotso, Pradhan, and Sukdeo [16], around 44% of ICT projects fail to satisfy company objectives, 43% of projects are delayed and under budget, and 18% of projects entirely fail owing to a lack of strategic efforts. Zadeh [35] mentioned that for every $1 billion invested in the United States, $122 million was squandered owing to poor project planning and performance.

To avoid these types of failures, the foremost duty of project managers is to provide an atmosphere where individuals can collaborate to accomplish a shared goal [28]. So, most of the issues that occur during a project encountered by project managers can be significantly reduced with the use of technology [28]. As a result, applying AI to project management can reduce the percentage of failed projects because AI will substantially alter the method in which they operate their businesses within the coming five years [32].

Predicted by 2030, AI-powered by big data, machine learning, and natural language processing will take care of 80% of project management responsibilities [29]. According to a recent PMI research, 44% of business projects in the sector are using resources and techniques supported by AI [29], and 70% of commercial organizations plan to begin cooperating with AI in project management [19]. AI can positively impact the domain of project management [25]. The tasks assigned to project managers can be easily automated, such as automatically sending messages to team members, gathering and examining past data, and making forecasts [28].

AI can be used in several areas within project management, including automation, planning sprints, evaluating team members, strategic partnerships, and chatbots. There are some challenges as well. Depending excessively on AI has the potential to weaken a person's critical thinking [28]. According to the report by Wang [36], three primary worries prevent AI from being widely utilized in project management, including confidentiality problems in AI-driven advertising, fears of losing one's job, and fears of complicated AI systems leading to failure and the lack of knowledge of using AI for project management.

3 Methodology

This study employed a mixed methods approach to investigate the impact of AI on project management in Norway. The research was conducted in two phases: a quantitative survey and qualitative interviews.

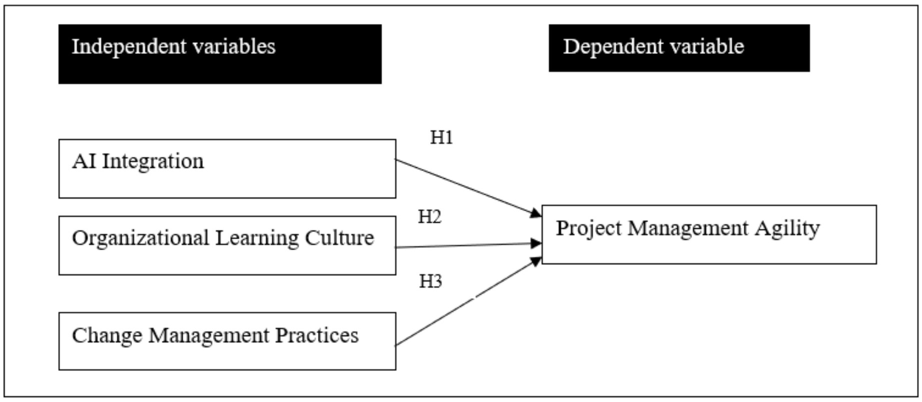

Fig. 1. Conceptual framework.

Based on the conceptual framework shown in Fig. 1, the following hypothesis has been formulated:

- H1: There is an impact of AI integration on project management agility in Norwegian companies.
- H2: There is an impact of organizational learning culture on project management agility in Norwegian companies.
- H3: There is an impact of change management practices on project management agility in Norwegian companies.

Project managers who are working in Norway have been taken as the sample, and a purposive sampling technique [26] has been used in this research.

In the first phase, a structured questionnaire was answered by 84 project managers across various industries. A sample was selected among persons on LinkedIn with the position listed as project manager located in Norway. Around 250 e-mails were sent to potential respondents. All e-mails were personalized. The initial e-mail was followed up with reminders for two weeks. The questionnaire comprised multiple-choice and Likert-scale questions designed to capture the extent of AI adoption, the specific AI tools used, and the perceived benefits and challenges of AI integration in project management.

The second phase involved conducting in-depth interviews with seven selected project managers who had significant experience with AI tools. These semi-structured interviews provided more profound insights into the practical applications of AI in project management, the challenges faced during implementation, and the overall impact on project efficiency and flexibility. The interviews were recorded, transcribed, and analyzed using thematic analysis [10] to identify common themes and patterns.

By combining quantitative data from the survey with qualitative insights from the interviews, the study aimed to provide a comprehensive understanding of how AI is transforming project management practices in Norway. This mixed-methods approach ensured a robust analysis, capturing both the breadth and depth of AI's impact in the field.

4 Findings and Discussion

This section presents the results from the quantitative analysis, followed by the results from the qualitative study.

4.1 Quantitative Analysis

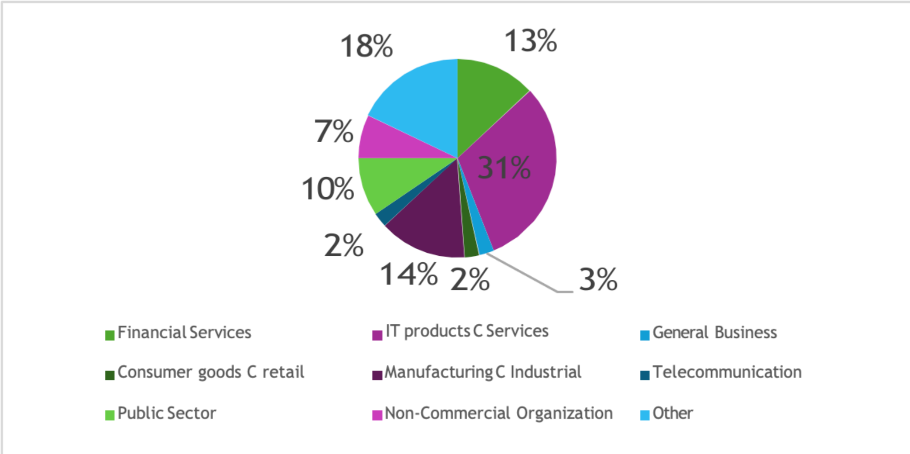

Fig. 2. Respondents by sector.

According to the findings, most project managers who are using AI are from IT products and services (31%). The breakdown of sectors is shown in Fig. 2.

Table 2. Reliability analysis.

Variable	Cronbach's Alpha	No. of items
AI Integration	0.90	7
Organizational Learning Culture	0.74	3
Change Management Practices	0.82	3
Project Management Agility	0.76	4

Reliability analysis has been conducted to find the internal consistency and reliability of the measuring scales [22] that are used to evaluate the variables under consideration, which include AI integration, organizational learning culture, and change management practices. The hypothetical Cronbach's Alpha values for organizational variables shown in Table 2 indicate generally strong reliability. Analysis indicates Cronbach's Alpha ranging from 0.74 to 0.90, which emphasizes acceptable levels for each variable. AI integration indicates Cronbach's Alpha value as 0.90, while organizational learning culture indicates 0.74, exhibiting good consistency. Project management agility indicates 0.76, which demonstrates a higher level of consistency throughout the variables. Change management practices indicate good reliability, which is 0.82 among the variables that are used to assess consistency. Accordingly, all variables display acceptable reliability.

These findings suggest that the measurement scales used to assess AI integration, organizational learning culture, change management practices, and project management agility are reliable and consistent in measuring the intended constructs within the sample population.

Table 3. Descriptive analysis.

Variable	N	Mean	Standard Deviation	Variance	Minimum	Maximum
AI integration	84	3.43	0.93	0.86	2.00	5.14
Organizational learning culture	84	3.07	1.22	1.49	1.33	5.67
Change management practices	84	3.22	0.84	0.70	1.33	5.33
Project management agility	84	3.94	0.94	0.88	1.50	5.25

The descriptive analysis shown in Table 3 provides an understanding of the characteristics within the study [7]. The variables were analyzed by using a sample of 84 project managers located in Norway.

The mean scores represent the average of each variable, AI integration, organizational learning culture, Change management practices, and project management agility, as 3.43, 3.07, 3.22, and 3.94. It demonstrates a medium to a higher level of AI adoption within the sample population, with an average AI integration value of 3.5. However, the range from 1.33 to 5.67 reflects substantial variability in AI adoption across different companies. This could be influenced by various independent variables, implying that AI adoption may enhance project management efficiency. In addition to this, values of standard deviation varied from 0.93 to 1.22, reflecting the level of dispersion variability of the scores around the mean.

Supportive organizational characteristics such as organizational learning culture, change management practices, and AI integration also show high mean values, suggesting that companies with solid internal practices are more likely to adopt AI effectively. Notably, the dependent variable, project management agility, has a slightly higher meaning, implying that AI adoption may enhance project management efficiency.

Pearson correlation coefficient is applied to determine the rate of entanglement of two characteristics that are in metric format [30]. The correlation coefficient, or 'r' with a range from -1 to 1, tells the type of relation between two descriptive variables. Table 4 showcases the correlation output generated through SPSS analysis.

Table 4. Correlational analysis.

Variable	PMA	AI-I	OLC	CMP
PMA	1	0.673**	0.755**	0.408**
AI-I		1	0.535**	0.219**
OLC			1	0.441**
CMP				1

Notes: **= Correlation is significant at the 0.01 level (2-tailed).
PMA – Project Management Agility
AI-I – AI Integration
OLC – Organizational Learning Culture
CMP – Change Management Practices

According to Table 4, there is a high degree of positive correlation between organizational learning culture with project management agility. There is a moderate degree of positive correlation between AI Integration and project management agility. It proves H1 (There is an impact of AI integration on project management agility in Norwegian companies), addressing the RQ1 (Does AI integration impact project management agility in Norwegian companies?). There is a low degree of positive correlation between Change Management Practices and project management agility by rejecting H3 (There is an impact of change management practices on project management agility in Norwegian companies) and addressing RQ3 (Does change management practices impact project management agility in Norwegian companies?).

Table 5. Regression analysis.

Predictor	Coefficient	Std. Error	t-Value	p-Value
Intercept		0.295	2.106	0.038
AI Integration	0.379	0.076	5.062	0.000
Organizational learning culture	0.508	0.062	6.231	0.000
Change management practices	0.101	0.079	1.435	0.155

Regression analysis is employed to assess how each of these independent variables affects the dependent variable of the project management agility. The results are shown in Table 5. Hence, it can be recognized which variables are vital for enhancing project management agility and how these variables interact comprehensively with each other. The p-value is less than the accepted beta value of 0.05. Therefore, it aligns with the outcomes specified for Hypothesis 1 (H1: There is an impact of AI integration on project management agility in Norwegian companies) and Hypothesis 2 (H2: There is an impact of organizational learning culture on project management agility in Norwegian companies) being accepted. Hypothesis 3 (H3: There is an impact of change management practices on project management agility in Norwegian companies) being rejected.

Considering the AI integration, the P value (0.000) is less than 0.05, and the beta value is 0.379. It is denoted that when the AI integration increases in one unit, it is expected that the project management agility will increase by 0.379. Substantially impacts the agility of project management practices, confirming the critical role of AI in enhancing operational efficiency and responsiveness. This acceptance of Hypothesis 1 underscores H1: There is an impact of AI integration on project management agility in Norwegian companies.

Furthermore, organizational learning culture also shows a significant positive impact (Beta value of 0.508 and p-value of 0.000), indicating that a learning-oriented organizational culture positively enhances the integration and effectiveness of AI in project management. This supports Hypothesis 2, highlighting that there is an impact of organizational learning culture on project management agility in Norwegian companies.

On the other hand, change management practices did not show a statistically significant effect (Beta value of 0.101 and p-value of 0.155), leading to the rejection of Hypothesis 3. This suggests that H3 does not significantly enhance the interaction between learning culture and AI integration in improving project management agility. Thereby, two hypotheses were accepted in the current research, and one was rejected.

Regression analysis provides insights into how AI integration is crucially supported by organizational learning, and it directly contributes to enhanced project management agility. However, change management practices in this scenario do not significantly influence this relationship, which may suggest a need for reevaluating the approach to change management in the context of AI deployment. Collectively, these results address research question 1 (RQ1: Does AI integration impact project management agility in Norwegian companies?) by confirming AI's significant impact on project management agility and providing indirect insights into research question 2 (RQ2: Does organizational learning culture impact project management agility in Norwegian companies?) suggesting that a supportive organizational culture likely fosters positive perceptions and interactions with AI among project team members and provide insights. Further, results offer insights regarding the third question (RQ03: Does change management practices impact project management agility in Norwegian companies?) relating to change management practices that need to be aligned as it is insignificant.

4.2 Qualitative Analysis

Thematic analysis has been done to examine the interview script [10], which investigates the integration and influence of AI technology in project management inside an organization.

AI integration in project management - Integration of AI within project management improves the effectiveness of project management activities [34]. Five interviewees said (Interviewees 2, 3, 4, 6, 7) that they (project managers) already use AI tools inside their company, and two (Interviewees 1 and 5) mentioned that they do not. The respondents used AI to automate their basic procedures, such as planning the project and allocating resources among team members. Those systems have the capability of monitoring real-time project updates. They mentioned that it gave them free time to work on other duties as well. Not only that, but it also increases precision in critical project decisions.

Impact of AI on efficiency and productivity - AI is making a significant contribution to the effectiveness of a project. Participants of the interview mentioned that AI can identify the timeline of the project and allocate the resources fairly. These are based on the needed data at the beginning of the integration. Two interviewees mentioned that it would make them more accessible because these technologies allow them to finish projects within the exact time frame and the estimated cost. Another important thing is that AI can provide better communication facilities.

Utilization of AI insights - AI utilization also talks about procedures for making decisions and managing the associated risks of the project. Some respondents mentioned that improvements need to be made in this area. They suggest that it would be better if there were some improvements to the risk prediction process.

Satisfaction with AI Integration - The satisfaction level of the AI use varied with respondents. Some mentioned that it has been very beneficial for them to perform their project management tasks, while some said that it is challenging because they are still trying to figure out the accuracy of the results. Two mentioned that they do not use AI for project management activities but would like to have it in the future if their organization allows it. Also, there are some suggestions from the respondents regarding the improvements in AI technologies and the accuracy of the outcome.

Innovation and new ideas - The receptiveness of the organization to new ideas and innovative approaches significantly influences the effectiveness of using AI in project management. The agility with which the organization incorporates new ideas correlates directly with its ability to stay competitive and effective in project delivery.

Change Management in AI Projects - Particularly regarding AI project integration, requires careful planning and communication [34]. Successful strategies include transparent communication, involving key stakeholders early in the process, and providing adequate support to address any resistance. Improvements could consist of more personalized communication strategies and greater involvement of team members in the planning stages to ensure smoother transitions [20].

5 Ethical Considerations

According to the study, respondents mentioned various ethical issues regarding the use of AI for project management. According to the respondents' feedback, bias is the main issue because the AI system is trained by feeding previous data and information, so it can be biased based on the data it already has when providing the output. It leads to unfair outcomes regarding the procedures of project management activities, such as dividing tasks among members and monitoring the individual performance of team members. So, respondents mentioned that there is a risk regarding expecting a highly accurate outcome. AI needs a large amount of data, including very private information. So, privacy issues and security [13] are other ethical issues mentioned by participants. Job replacement is also another issue. Most noted that if companies started to use AI for everything, there would be less need for staffing. So, there may be a smaller number of job vacancies for lower-level employees.

6 Limitations and Future Work

The small sample size of 84 project managers is the main limitation of this study, and it will restrict the generalizability [9]. Also, the sample from which the data was collected was from Norway.

Future research could build on the collected data to make a longitudinal study. The research could also be expanded to more geographical regions. Finally, it would be possible to elaborate upon some of the cases from the qualitative research to analyze in more depth how tools are used in practice.

The advancement of cutting-edge technology regarding AI in project management may cause less relentless findings. Future studies should be more concerned with technological advancements.

7 Conclusion

The integration of AI in project management is transforming how projects are managed in Norway, offering significant improvements in efficiency and flexibility. This study's mixed-methods approach revealed that many project managers have already adopted AI tools, using them for task automation, predictive analytics, and resource management. AI's ability to streamline routine tasks and provide valuable insights through data analysis enables project managers to make more informed decisions and adapt to changing project demands. Despite some challenges, such as the learning curve and data security concerns, the overall positive reception highlights AI's potential to revolutionize project management. As AI technology continues to advance, its role in project management is expected to grow, further enhancing project outcomes. This study underscores the importance of ongoing research and the development of best practices for AI adoption in project management to harness its capabilities.

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